

Bare noun interpretation in Russian requires syntax

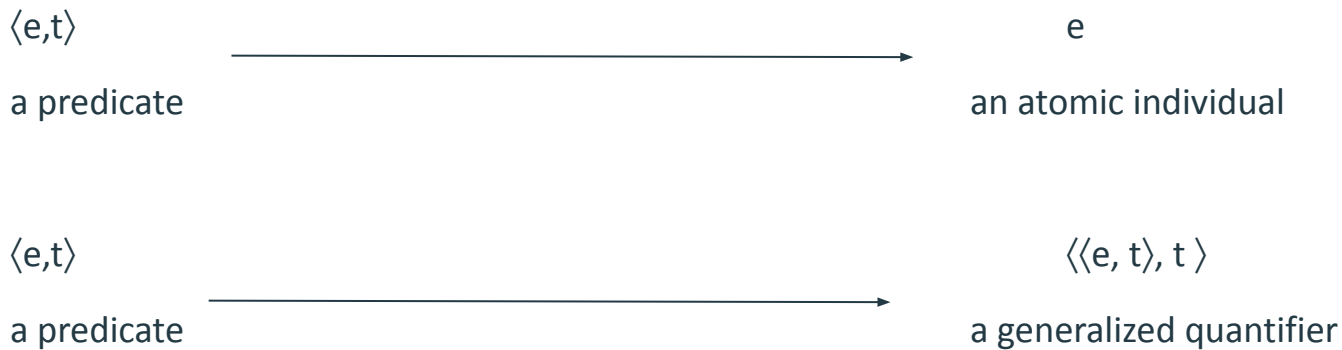
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big picture

how do predicates become arguments?



making an argument

Option 1: Existential quantification

$\lambda P. \lambda Q. \exists x [P(x) \wedge Q(x)]$

turns $\langle e, t \rangle$ into $\langle \langle e, t \rangle, t \rangle$

$\llbracket \text{A girl entered the room} \rrbracket = \exists x. \text{girl}(x) \wedge \text{entered-the-room}(x)$

making an argument

Option 1: Existential quantification

Option 2: *iota* operation

$\lambda P. \iota P_s$, if there exists a unique maximal entity in P, undefined otherwise turns $\langle e, t \rangle$ into e

$\llbracket \textit{The girl sat down} \rrbracket = \text{sat-down}(\iota x. \text{girl}(x))$

making an argument

Option 1: Existential quantification

Option 2: *iota* operation

Option 3: *nom* operation (kind formation; $\cap$)

a function from indices to the maximal entity that is the extension of the predicate at the index;

only applies to kind-level predicates

turns $\langle e, t \rangle$ into e (the corresponding kind)

$\llbracket \textit{Dinosaurs are extinct} \rrbracket = \text{be-extinct}(\cap \text{dinosaurs})$

languages with vs. without articles

- In languages with articles, $\exists$ and *iota* are lexicalized (e.g. as *the* and *a/an* in English)
- In languages without articles, like Russian, both “indefinite” and “definite” readings are available

one can introduce an individual using a bare noun

V komnatu zašli devočka i mal'čik. Devočka sela na stul.
in room entered.PL girl and boy girl sat on chair
'A boy and a girl entered the room. The girl sat on the chair.'

and refer to an already introduced individual using a bare noun

core empirical question

Are bare singulars really semantically ambiguous between definite and indefinite, or does syntax (or something else) quietly constrain them?



theoretical landscape

three approaches to bare singulars

1. Ambiguity Approach

Bare singulars start out as properties and then freely shift to either definite or indefinite.

Inspired by Krifka's (2003) work on bare plurals; Liu et al. (2023), LeBruyn et al. (2025) for bare singulars

2. Indefinite-Based Approach

Bare singulars are, in fact, indefinite. Formalisms differ, but generally suggest ambiguity between existential quantification and choice functions.

Inspired by Heim (2011); Šimík (2019), Šimík & Burianová (2020), Šimík & Demian (2020), Seres & Borik (2021)

3. Kinds Approach (Neo-Carlsonian Approach)

Bare singulars are, in fact, definite. They have to be unique and they have the scopal properties of definites.

Dayal (2004, 2011, 2018)

decomposing definiteness

Uniqueness	Familiarity
There is a single relevant individual in a contextually restricted domain that satisfies the description	The referent is discourse-given or salient, e.g. by prior mention, situational anchoring, or shared knowledge

- These components need not covary
- Successful reference can be licensed by familiarity even when uniqueness is violated
- Experimental results that do not manipulate these factors independently are theoretically underdetermined (Srinivas 2021)

adding complexity: word order and information structure

- Russian is a language with flexible word order shaped by information structure
- Geist 2010: bare singulars can be indefinite only in the comment (=not topic)
- Preverbal position correlates with topicality
- Postverbal position licenses weak existential readings

preverbal subject - Geist reports it to be obligatorily definite (anaphoric)

a. **Devočka** voš-l-a v dom.
girl enter-PST-FEM in house
'The girl/*A girl entered the house.'

b. V dom voš-l-a **devočka.**
in house enter-PST-FEM girl
'A girl/The girl entered the house.'

postverbal subject - either interpretation is possible

previous experimental result: no uniqueness?

- Šimik & Demian (2020) tested whether Russian bare singulars require a unique referent
- Other than the uniqueness of the referent, they manipulated:
 - word order (SV vs. VS)
 - prosody (SV with unmarked S vs. SV with prosodically marked S)
- Task: covered box paradigm. Participants were given a picture supporting or not supporting a unique referent + a sentence; what was measured was whether they indicate that they'd prefer some different picture for the given sentence



Šimik & Demian's (2020) result and conclusion:

No interaction between uniqueness and word order and no interaction between uniqueness and prosodic marking

No uniqueness effect overall

→ Bare singulars are presupposition-free and existential.

why null results are underdetermined

- A null effect is compatible with multiple theories
- Possible explanations provided by the authors include:
 - Word order manipulation failed to be interpreted by the participants as manipulation of topicality
 - Word order manipulation do correspond to definiteness manipulation, just uniqueness remains unaffected by it; meaning that other components of meaning, such as familiarity, identifiability, non-ambiguity, or salience are more important for licensing definites
 - The weakness of the design overall



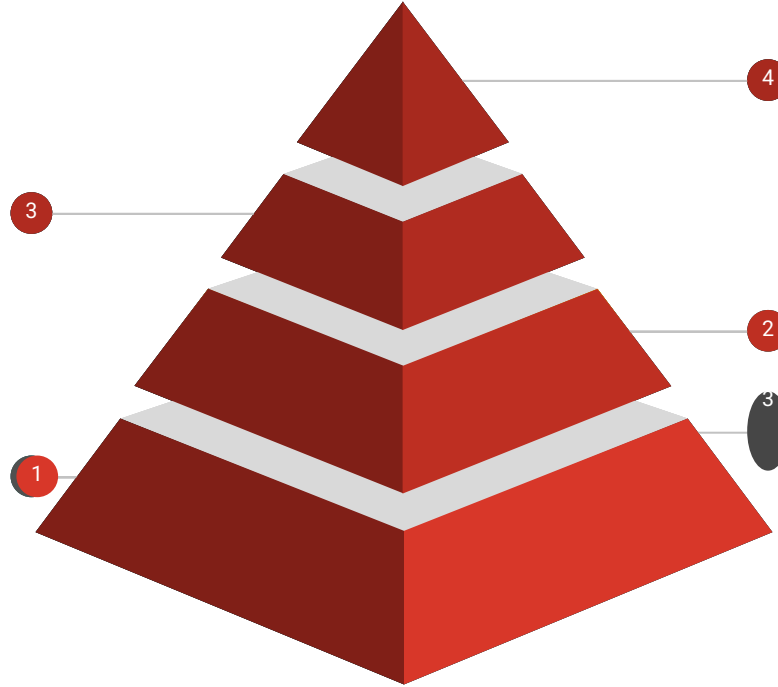
the current study:
progressive refinement

Vestibulum congue tempus

Lorem ipsum dolor sit amet, consectetur
 adipiscing elit, sed do eiusmod tempor.
 Donec facilisis lacus eget mauris.

Vestibulum congue tempus

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strategy: word order as a controlled cue

- Word order in Russian is tightly linked to information structure, rather than grammatical roles
- Preverbal bare singulars bias towards discourse-given
- This does not mean that word order encodes definiteness or uniqueness directly
- Instead, it likely correlates with familiarity-related properties (aboutness, givenness, domain anchoring), which can in turn mask or satisfy uniqueness requirements

Strategy in this study

Use word order as a controlled cue to familiarity, in order to test uniqueness while holding familiarity approximately constant

baseline diagnostic: familiarity and word order

- 2 X 2 forced binary-choice task
- Tests whether preverbal bare singulars tend to be interpreted as familiar more so than postverbal bare singulars
- Presentational contexts with intonational pattern kept constant as neutral narrative

Unfamiliar condition

Lena and Tom don't have pets. Lena calls Tom and hears barking in the background.

Tom looks outside and sees that a group of street dogs ran into the backyard.

Lena: What is going on?

Which of Tom's answers would sound best?

Familiar condition

Lena and Tom have a dog named Max, who barks whenever someone walks by. Lena calls and hears barking. It's Max, who's out in the yard.

Lena: What is going on?

Which of Tom's answers would sound best?

a. **Sobaka** laet vo dvore.
dog barks in backyard

b. Vo dvore laet **sobaka**.
in backyard barks dog

presented in audio format,
recorded and post-processed
to keep intonation neutral;
randomized

counterbalanced across
experimental lists so that no
participant sees both
versions of the same context

+ Familiar
SV

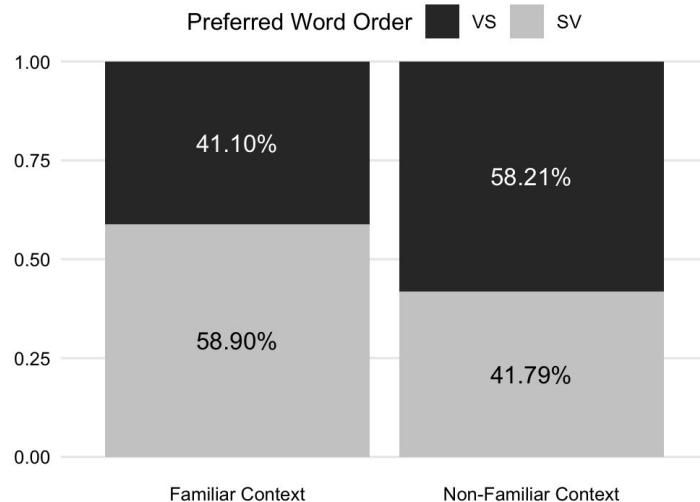
+ Familiar
VS

- Familiar
SV

- Familiar
VS

baseline results

- When the referent is familiar, participants tend to prefer to refer to it with a preverbal NP; when it's unfamiliar, they switch to the inverted word order
- Analyzed using linear mixed-effect regression predicting word order choice from familiarity with random intercepts for participants and experimental items
- Effect significant, $\beta = -0.85$, $SE = 0.16$, $p < .001$; moderate-to-high effect size ($OR = 2.34$)
- Meaningful variation across participants ($N = 128$) and experimental items ($n = 16$), but no clear pattern in the distribution of either of the random effects



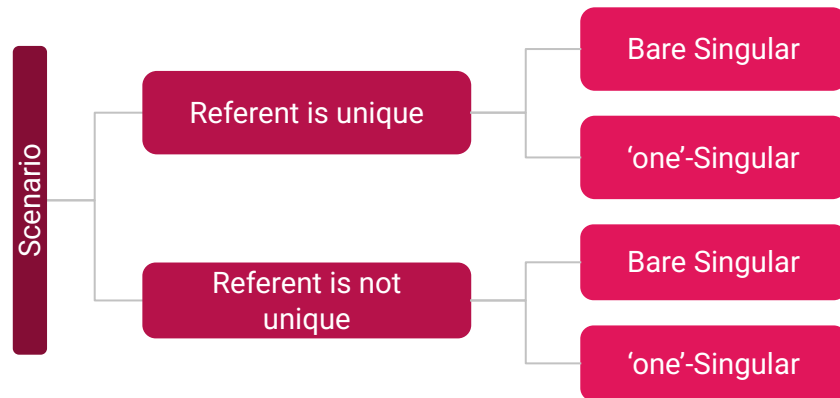
Word order correlates with familiarity – even in presentational contexts

exp A and B

uniqueness in presence vs. absence of familiarity

Do Russian bare singulars show sensitivity to uniqueness when familiarity is held constant?

- Experiment A: do bare singulars presuppose uniqueness when familiar?
 - In this experiment, we tested SV
- Experiment B: do bare singulars presuppose uniqueness when non-familiar?
 - In this experiment, we tested VS
- Control condition: singulars modified by the numeral *odin* 'one'
- 2 X 2 factorial design: acceptability judgment task with joint presentation
- Each task had 60 participants and 16 experimental items + 24 filler tasks



exp A

testing uniqueness when familiarity is supported

the individual(s) in the scenario were presented as clearly familiar

Unique condition

Masha and Andrey have a pet hamster named Pyatachok. He lives in a cage in the living room. One day, Masha and Andrey were having dinner. Masha went to the room to get napkins and saw that Pyatachok was gone, and the cage door was open. Masha screamed.

Andrey asked from the kitchen: “What happened?”

How appropriate is each of these possible replies from Masha in this situation?

- a. **Xomjak** sbežal iz kletki.
hamster ran.away from cage
'The hamster ran away from the cage.'
- b. **Odin xomjak** sbežal iz kletki.
one hamster ran.away from cage
'One hamster ran away from the cage.'

answers were presented as audios
with controlled neutral prosody;
both options were presented
jointly with an unlabeled scale for a
rating (translated into 1-100)

in both rated options, the
target NP was a preverbal
subject



Насколько каждый из этих ответов подходит в этой ситуации?

- 1.  0:00 / 0:01
- 2.  0:00 / 0:01

Совсем неподходящий ответ

Отлично подходит как ответ на этот вопрос

Аудио 1



Аудио 2



exp A

testing uniqueness when familiarity is supported

Non-unique condition

Masha and Andrey have two pet hamsters, Busya and Pyatachok. They live in a cage in the living room. One day, Masha and Andrey were having dinner. Masha went to the room to get napkins and saw that Pyatachok was sitting in the cage alone. Busya was gone, and the cage door was open. Masha screamed.

Andrey asked from the kitchen: "What happened?"

How appropriate is each of these possible replies from Masha in this situation?

- a. **Xomjak** sbežal iz kletki.
hamster ran.away from cage
'The hamster ran away from the cage.'
- b. **Odin xomjak** sbežal iz kletki.
one hamster ran.away from cage
'One hamster ran away from the cage.'

in the non-unique condition, two
equally salient individuals were
explicitly introduced in the scenario

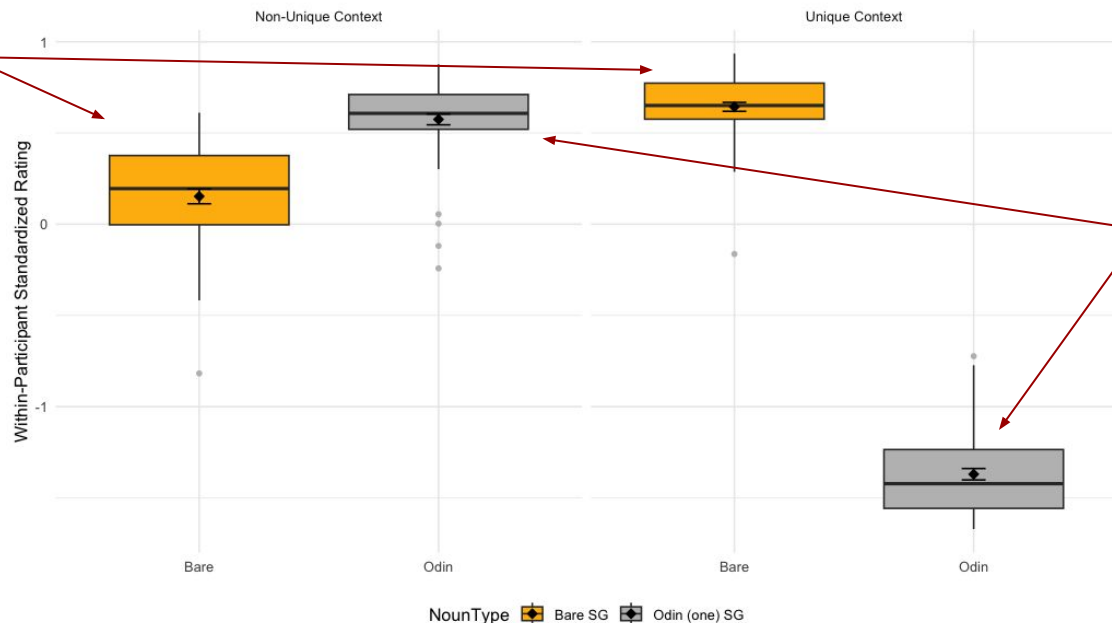


exp A: uniqueness effect emerges

bare singulars are rated significantly higher in unique contexts, compared to non-unique

on average, familiar bare singulars were rated 0.38 SD higher when unique

this effect size is medium-to-large (Cohen's $D=0.67$)



strong interaction effect of Uniqueness and NounType ($\beta=2.31$, $p<.001$)

the effect of uniqueness manipulation on the control condition ('one'-marked nouns) is opposite

exp B

testing uniqueness when familiarity is suppressed

Unique condition

Mom is watching the news. Today's news is about a zoo in a small town. The report says that this morning, a staff member came to feed a raccoon — the only one in this zoo — and saw that the cage was open and the raccoon was gone.

Dad asks: "What did they say on the news?"

How appropriate is each of these possible replies from Mom in this situation?

- a. Iz odnogo zooparka sbežal enot.
from one zoo ran.away raccoon
'A raccoon/The raccoon ran away from a zoo.'
- b. Iz odnogo zooparka sbežal odin enot.
from one zoo ran.away one raccoon
'A raccoon/One raccoon ran away from a zoo.'

in this task, the target NPs
were postverbal subjects



exp B

testing uniqueness when familiarity is suppressed

Non-Unique condition

Mom is watching the news. Today's news is about a zoo in a small town. The report says that this morning, a staff member came to feed two raccoons and saw that the cage was open and one of the raccoons was gone. The other was still sleeping in its little house.

Dad asks: "What did they say on the news?"

How appropriate is each of these possible replies from Mom in this situation?

- a. Iz odnogo zooparka sbežal enot.
from one zoo ran.away raccoon
'A raccoon/The raccoon ran away from a zoo.'
- b. Iz odnogo zooparka sbežal odin enot.
from one zoo ran.away one raccoon
'A raccoon/One raccoon ran away from a zoo.'

options presented jointly as audio
with controlled prosody, randomized

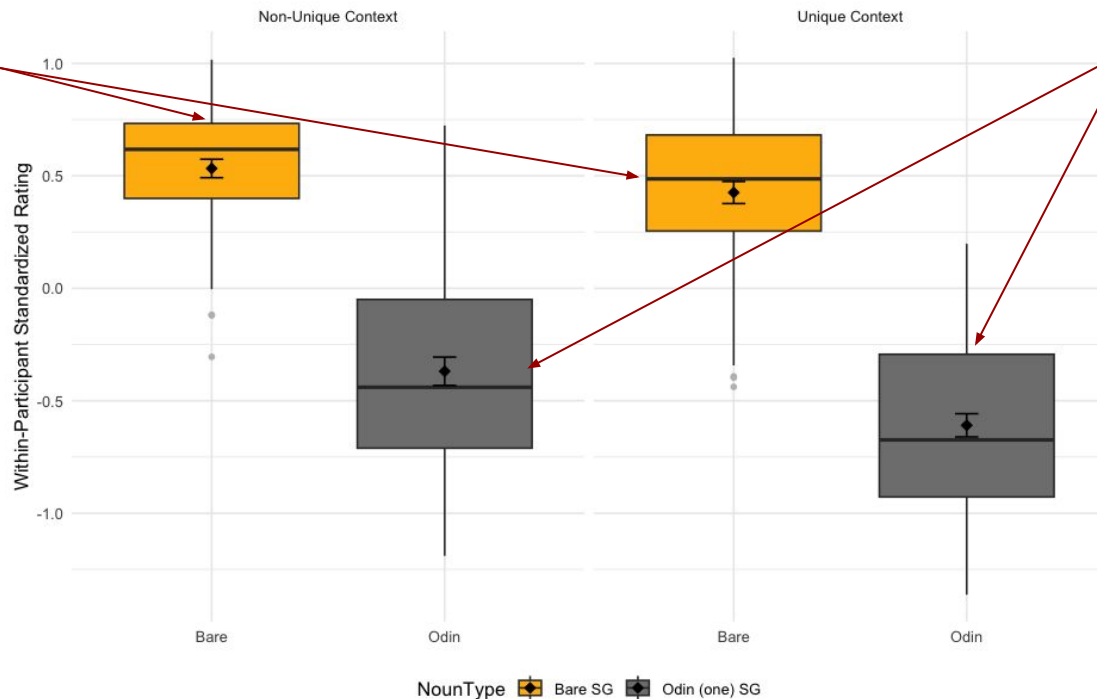
again, two equally salient individuals were introduced in the non-unique condition; non-familiarity was also enforced by using clearly unfamiliar setting (*from some zoo*)



exp B: no visible uniqueness effect

no significant difference in acceptability of bare singulars across unique vs. non-unique contexts ($p=0.24$)

the task contained fillers used to filter for adequate answers → the null effect does not come from participant inattentiveness



no interaction effect ($p=0.5$)

low ratings for postverbal 'one'-nouns; potential incompatibility with the postverbal position?

interim puzzle: when does uniqueness disappear?

Summary of the results so far:

- When familiarity is supported by the context and the bare singular subject is preverbal,
non-uniqueness of the referent reduces acceptability
- When familiarity is suppressed by the context and the bare singular subject is postverbal
non-uniqueness of the referent does not reduce acceptability

**If bare singulars are in principle sensitive to uniqueness,
why is this sensitivity visible in one configuration but not the other?**

In particular, why does the absence of familiarity not automatically make non-uniqueness detectable?

Hypothesis:

Bare singulars require uniqueness, but information structure can restrict the relevant domain so that uniqueness is satisfied trivially.

Key ingredients:

- Uniqueness is interpreted relative to a domain, which can be contextually restricted
- Information and sentence structure can further restrict the domain to a singleton
- The postverbal position is typically associated with non-topic, especially in sentences with inversion

Consequences:

- In Experiment B, non-unique discourse contexts may have nevertheless yielded unique domains, making violations of uniqueness empirically invisible

exp C: incompatible predicates

- The context with incompatible predicates makes the domain restriction impossible
- Keeping the postverbal position
- Triggering a truly topicless, presentational context



the sentences with bare singulars use the postverbal position to make it comparable to exp B

-Uniqueness

Masha looks into the toy and sees two dogs. Liza asks: "What do you see there?"

- Na kartinke spit **sobaka** i begaet po krugu **sobaka**.
in picture sleeps dog and runs in circles dog
- Na kartinke **odna sobaka** spit i **odna sobaka** begaet po krugu .
in picture one dog sleeps and one dog runs in circles
Intended: 'In the picture, a dog is sleeping and a dog is running in circles'.

the sentences with 'one'- singulars use the preverbal position: pilot testing showed that postverbal 'one' is degraded for independent reason, and we are interested in a stable acceptable control

exp C: incompatible predicates

- The context with incompatible predicates makes the domain restriction impossible
- Keeping the postverbal position
- Triggering a truly topicless, presentational context

in the unique condition, there are
two different individuals and
therefore a unique dog



+Uniqueness

Masha looks into the toy and sees a dog and a bunny. Liza asks: "What do you see there?"

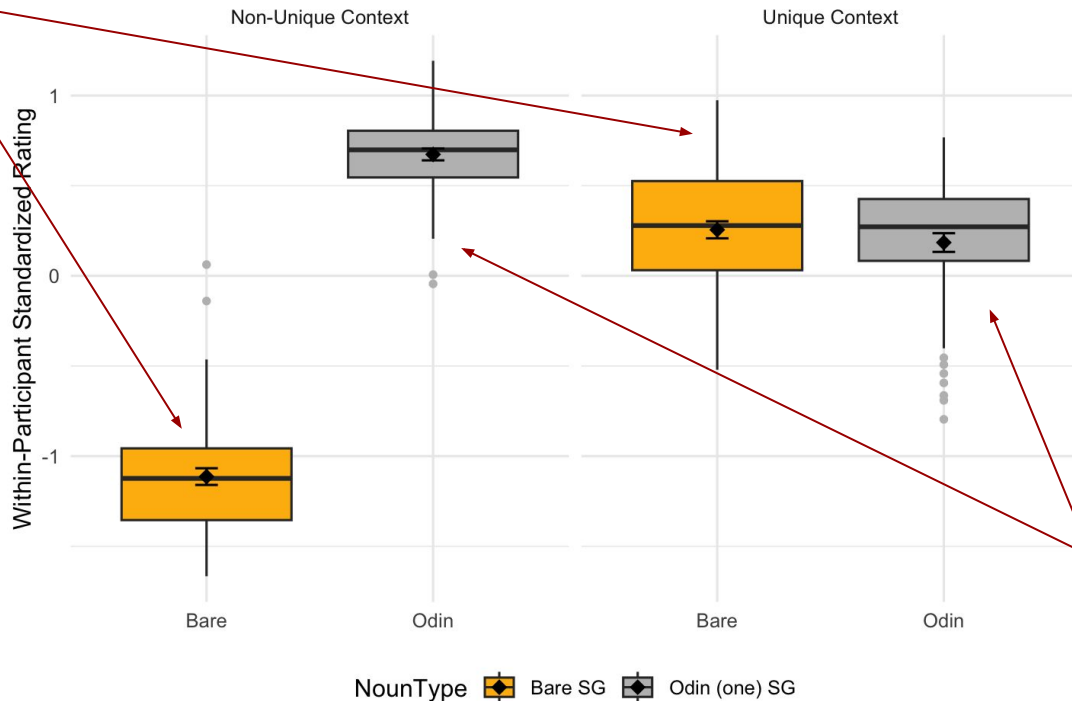
- Na kartinke spit **sobaka** i begaet po krugu **zajčik**.
in picture sleeps dog and runs in circles bunny
- Na kartinke **odna sobaka** spit i **odin zajčik** begaet po krugu.
in picture one dog sleeps and one bunny runs in circles
Intended: 'In the picture, the dog is sleeping and the bunny is running in circles'.

- Acceptability judgment task with joint presentation; 16 experimental items, 2:3 experimental/filler ratio, 60 participants recruited

exp C: uniqueness effect visible again

bare singulars largely degraded when non-uniqueness is forced ($\beta = -1.25$, $p < .001$; Cohen's $D = 1.8$)

strong interaction effect ($\beta = -1.7$, $p < .001$)



opposite effect in 'one'-marked singulars; significant but of smaller magnitude ($\beta = 0.45$, $p < .001$)



takeaway

what we learned

0. A validated diagnostic for controlling familiarity

Word order reliably correlates with familiarity in our results. This allows us to use it to help keep familiarity constant in subsequent experiments.

1. Bare singulars exhibit systematic sensitivity to uniqueness

Across experiments, bare singulars are penalized when the context forces reference to more than one equally salient individual, provided that trivial satisfaction of uniqueness through domain restriction is blocked.

2. Apparent counterexamples reflect domain restriction, not existential meaning

Postverbal configurations and information structure can restrict the domain of interpretation so that uniqueness is satisfied vacuously; this does not constitute evidence of existential semantics.

3. Syntax matters at the interface

Syntactic position shapes information structure and domain restriction, determining whether uniqueness requirements are empirically visible.

conclusion: why null effects do not settle the debate

Against ambiguity-based approaches

Our data do not support lexical ambiguity between definite and existential interpretations: bare singulars do not show mixed or context-free existential behavior.

Against indefinite-based approaches

We see no genuine existential behavior. Occasional null effects do not motivate an existential analysis, as they can be explained by trivial satisfaction of uniqueness under domain restriction.

Bare singular interpretation in Russian requires uniqueness → iota-like interpretation rather than full-fledged $\exists$ -quantification

Syntactic position contributes to information structure, thereby determining whether the uniqueness requirement is empirically visible

acknowledgments and references

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- Dayal. (2004). Number marking and (in)definiteness in kind terms. *Linguistics and Philosophy*, 27, 393–450.
- Dayal (2011). Hindi pseudo-incorporation. *Natural Language & Linguistic Theory*, 29(1), 123–167.
- Dayal (2018). (In)definiteness without articles: Diagnosis, analysis, implications. In Sharma & Bhatt (Eds.), *Trends in Hindi linguistics* (pp. 1–26). Walter de Gruyter.
- Geist, L. (2010). Bare singular NPs in argument positions: Restrictions on indefiniteness. *International Review of Pragmatics*, 2, 191–227.
- Heim, I. (2011). Definiteness and indefiniteness. In K. von Stechow, C. Maienborn, & P. Portner (Eds.), *Semantics: An international handbook of natural language meaning* (Vol. 2, pp. 996–1025). De Gruyter.
- Seres, D., & Borik, O. (2021). Definiteness in the absence of uniqueness: The case of Russian. In *Proceedings* (pp. 339–363). Language Science Press.
- Seres, D., Borràs-Comes, J., & Espinal, M. T. (2021). Uniqueness in languages with and without articles: Catalan vs. Russian. *Beyond Philology*, 18(3), 163–195.
- Šimík, R., & Burianová, M. (2020). Definiteness of bare NPs as a function of clausal position: A corpus study of Czech. In T. Ionin & J. E. MacDonald (Eds.), *Formal Approaches to Slavic Linguistics 26: The Urbana-Champaign Meeting 2017* (pp. 343–361). Michigan Slavic Publications.
- Šimík, R., & Demian, C. (2020). Definiteness, uniqueness, and maximality in languages with and without articles. *Journal of Semantics*, 37(1), 311–366.
- Šimík, R., & Wierzbica, M. (2015). The role of givenness, presupposition, and prosody in Czech word order. *Semantics and Pragmatics*, 8(3), 1–103.
- Srinivas, S. (2021). *The semantics of (in)definiteness in bare vs. non-bare nominals: A study of Kannada and English* (Doctoral dissertation). Johns Hopkins University.